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Heavy rainfall following a summer drought stimulates soil redox dynamics and facilitates rapid and deep translocation of glyphosate in floodplain soils†

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We present field data on the effects of heavy rainfall after drought on the mobility of glyphosate and redox conditions in a clayey floodplain soil. By applying glyphosate together with deuterated water as conservative tracer in combination with time resolved *in situ* redox potential measurements, the spatial and temporal patterns of water infiltration and pesticide transport as well as the concomitant changes of the redox conditions were revealed. Our findings demonstrate that shrinkage cracks in dry soils can serve as effective transport paths for atmospheric oxygen, water and glyphosate. The rain intensity of a typical summer storm event (approx. 25 mm within one hour) was sufficient to translocate deuterated water and glyphosate to the subsoil (50 cm) within 2 hours. Soil wetting induced partial closure of the shrinkage cracks and stimulated microbial activity resulting in pronounced dynamics of *in situ* soil redox conditions. Redox potentials in 40 to 50 cm depth dropped permanently to strongly reducing conditions within hours to days but fluctuated between reducing and oxidizing conditions in 10 to 30 cm depth. Our findings highlight the close link between the presence of macropores (shrinkage cracks), heavy rainfall after drought, redox dynamics and pesticide translocation to the subsoil and thus call for further studies addressing the effects of dynamic redox conditions as a limiting factor for glyphosate degradation.

Environmental significance

This field study highlights the role of shrinkage cracks in dry floodplain soils for the redox dynamics in the soil profile and the translocation of strongly sorbing pesticides to the subsoil. The precipitation of a typical summer rainstorm was sufficient to transfer previously applied glyphosate rapidly to the subsoil and to induce a decrease of redox potentials in the subsoil by cutting off the oxygen supply through macropores. Strongly reducing conditions potentially impair further biodegradation of glyphosate in the subsoil. As drought and heavy rain events are expected to increase with progressing climate change, the significance of transient soil features such as shrinkage cracks for physical and biogeochemical processes of soils will also increase.

Introduction

Floodplains are landscape elements of high biogeochemical activity and turnover of natural and anthropogenic compounds.[1,2] Periodic changes of the water regime induce variations of redox conditions, of the availability of electron acceptors and donors (*e.g.*, oxygen, iron) and thus the sequestration or remobilization and turnover of redox inert (*e.g.*, phosphorus)[3] and redox sensitive nutrients (*e.g.*, pentavalent nitrogen in nitrate) and pollutants.[1] Floodplain soils in mid-latitudes show pronounced seasonal water dynamics with waterlogged and oxygen-depleted conditions throughout the cold season and extended periods of unsaturated and oxic conditions during summer.[4,5] Under waterlogged conditions, long residence times for both percolating water and solutes are

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expected. During and after drought periods in summer and fall, gas exchange with the atmosphere and transport of matter from the surface to the subsoil is facilitated by the formation of shrinkage cracks. Due to climate change with increasing incidents of extreme droughts and precipitation,[6–8] the switch between dry and waterlogged conditions will occur more frequently and with shorter periods also within a season. This will impact not only the water regime in soils but likely also the mobility and transformation of nutrients and agrochemicals in the pedosphere.[9,10] The impacts of a rapid wetting after drought due to heavy rain on pesticide transport in floodplain soils, however, have rarely been studied and are not well understood.

Macropores such as shrinkage cracks may enhance rapid and irregular infiltration of water, solutes, and particles[11,12] as well as gas exchange with the atmosphere. Factors relevant for the occurrence of preferential flow in macropores include pore size, pore continuity and tortuosity[13–17] as well as intensity and quantity of irrigation or precipitation.[18] Dry soils initially exhibit a low capacity to take up water which favors water flow in macropores.[12,19,20]

Transport along preferential flow paths can be an important pathway for pesticide mobility.[21–24] As pesticides largely differ in their physico-chemical properties such as water solubility and affinity to adsorb to the surface of soil mineral particles,[25] their susceptibility to preferential flow also varies.[26] Due to its strong sorption to soil mineral particles,[27–29] glyphosate was long regarded to be fairly immobile in top soil and little susceptible towards leaching to the subsurface.[30] Nevertheless, significant leaching of glyphosate from soils to tile drains was reported[31–35] and the question arises if and under which conditions macropores such as shrinkage cracks facilitate its rapid transport to the subsoil.

As for nutrients and pesticides, also oxygen transport is facilitated through air-filled macropores,[36] a process which largely impacts the type and dynamics of biogeochemical conditions in fine-grained soils. Rapid oxygen transport through shrinkage cracks to the subsoil under dry conditions, however, can be terminated by precipitation as shrinkage cracks close due to swelling of clays. In combination with aerobic microbial activity this triggers oxygen consumption, thus lowering soil redox state especially in loamy floodplain soils rich in organic carbon. The presence or absence of molecular oxygen is a key factor for the functioning of microbial communities carrying the metabolic potential for pesticide degradation.[37]

Beside molecular oxygen, floodplain sediments contain alternative electron acceptors for microbial respiration such as iron(oxy)hydroxides. Alternating redox conditions in such soils are often signified by redoximorphic marks in the solid matrix caused by precipitation of redox-active iron (Fe) minerals.[38] In contrast to iron(oxy)hydroxides, iron containing clay minerals and soil organic matter (SOM) have only recently come into focus as redox-sensitive soil components.[39–41] They are capable of reversibly accepting and releasing electrons without leaving visible redoximorphic features and without dissolution or the formation of secondary phases.[42–44] Furthermore, SOM and Fe-species are highly redox-active towards commonly used redox probes.[45,46] Thus, in the presence of such redox sensitive phases, field monitoring of redox potentials (E_H) provides meaningful information on the redox state and the redox dynamics despite the well-known problems of interpreting measured E_H values due to mixed potential formation and redox disequilibria.[47–49]

We hypothesize that (i) shrinkage cracks are preferential pathways for water and for glyphosate (as a model compound for strongly sorbing pesticides) leaching into the subsoil, and (ii) their closure during wetting of the soil controls oxygen supply and thus the dynamics of redox potentials. As biodegradation of glyphosate is strongly impaired under anoxic conditions,[50,51] it is important to understand the interplay of shrinkage crack dynamics, oxygen supply and glyphosate leaching to assess the fate of glyphosate in subsoils.

We applied glyphosate to a dry Gleysol exhibiting a network of shrinkage cracks that developed after summer drought periods. We simulated a heavy rain event with irrigation water that was labelled using deuterated water (D_2O) as conservative tracer to quantitatively follow water infiltration. Glyphosate concentrations and the hydrogen isotope ratios (δ^2H) of soil pore water were determined along soil cores (50 cm) taken at spots with and without shrinkage cracks present at the soil surface. Changes of redox conditions indicating the dynamics of oxygen supply and consumption were recorded *in situ* by means of permanently installed and spatially resolved sensors. We present data on soil water dynamics together with glyphosate concentrations along soil cores immediately after irrigation and the response of *in situ* redox measurement to the simulated precipitation for 14 days.

Materials and methods

Field site

The field site was in the Ammer valley floodplain approx. 7 km west of Tübingen (TÜ), SW Germany (lat. 48.521266, long. 8.972687, 394 m a.s.l.). The Ammer catchment is located in the South–Western German Escarpment within the geological units of the Middle and Upper Triassic. The west of the floodplain is dominated by limestones from the Muschelkalk units, the east lies within Keuper units mainly comprising terrestrial sediments such as sand-, mud- and dolostones including gypsum.[52–55]

The floodplain with the field test site is in the eastern part of the catchment. It is stretched in a west–east direction and is enclosed by Keuper hills on its northern and southern rims. The hydrostratigraphy comprises (from top to bottom) a regional aquitard (Upper Clay Aquitard), a confined aquifer made of tufa sediments (Tufa Aquifer), another regional aquitard (Lower Clay Aquitard) and another aquifer made of mainly gravel and clay sediments (Gravel Aquifer).[54] The Upper Clay Aquitard, which comprises the investigated soil, consists of fine-grained clay mineral-rich alluvial sediments from the surrounding Keuper hills.[56,57] On average it is 2 m thick and water-saturated in its lower half throughout the year.[54] The soil is characterized as Gleysol[57] and some cores from the study from Martin *et al.*[54] showed typical gleyic properties (personal communication with S. Martin), *i.e.*, color-distinguishable oxidized and reduced horizons[58] (Fig. S1, ESI, Section S1†). With the cores (50 cm) taken for the presented study, only the Ah and the Go horizons were sampled.

The soil has a uniform texture of 11 ± 4 wt% of sand, 56 ± 6 wt% of silt and 32 ± 4 wt% of clay over the first 50 cm of depth (Fig. S2,† determined following DIN EN ISO 17892-4,[59] detailed protocol see ESI Section S2†) and a soil organic matter (SOM) content (quantified as total organic carbon (TOC)) of 2–2.5 wt% in the upper 15 cm and around 1 wt% between 20 and 50 cm depth (Fig. S3,† for details on the methodology see ESI Section S3†). The soil pH varied from 7.0 to 7.3. It was measured on six samples, two each from depth segments 0–10 cm, 20–30 cm and 40–50 cm from irrigated plots. The pH was determined after 30 min of stirring in a 1 : 1 (v : w) mix of deionized water and soil.

We performed the experiment on a plot of arable land immediately after the harvest of summer barley during summer 2019 (more details in Sections "meteorological conditions" and "plot treatment"). Information on prior use and treatment comes from personal communication with the farmer: during winter 2018/19, a catch crop (radish and mustard) was cultivated on the agricultural field and in February 2019 after frost, it was incorporated 10 cm deep with the cultivator. Additionally, during winter 2018/19, the field was fertilized with horse manure (20 t ha^{-1}), and in spring 2019, 300 kg ha^{-1} of calcium ammonium nitrate fertilizer (27% N) was applied. During the cultivation of summer barley, the field was treated with the herbicide Ariane-C (active ingredients: 100 g L^{-1} Fluroxypyr (as Methylheptylester 144 g L^{-1}), 80 g L^{-1} Clopyralid + 2.5 g L^{-1} Florasulam). The field had not been treated with glyphosate for 10 years before our study. Residues of glyphosate from previous applications or spray drift were not detected in prior samples.

Meteorological conditions

The experimental site lies in a region of temperate oceanic climate with a long-term mean precipitation of 720.9 mm $year^{-1}$ and a mean annual temperature of 9.4 °C.[60] Weather data (hourly precipitation and temperature, Fig. S4†) was retrieved from a station in TÜ-Unterjesingen (lat. 48.52950, long. 8.98380, 439 m a.s.l., which is 45 m higher than the actual experimental site).[61] The experiment took place from 15th July to 30th July 2019. The two-weeks period before the experiment was dominated by maximum temperatures above 20 °C on all but two days and rainfall events on three days (9.1, 5.1, and 33.1 mm, respectively). All of them were of moderate rain intensity according to the classification of the American Meteorological Society.[62] During the experiment, minimum temperature was 10.2 °C during the night (night average 17.7 °C) and maximum temperature was 36.3 °C during daytime (day average 24.5 °C). Reference evapotranspiration (ETo) for the whole month July 2019 was calculated as 4.2 mm day^{-1} with the FAO ETo calculator following the FAO guideline[63] including data on temperature (in °C, monthly min, max, and mean), relative humidity (in %, monthly min, max, and mean), and solar radiation (in Wh m^{-2}, monthly mean) (ESI Section S5, Table S1†). Between 15th and 31st July, we recorded 12 days with T_{max} > 25 °C, from which 6 had T_{max} > 30 °C. Throughout the experiment, only two natural rain fall events occurred with a short rain fall event on 21st July 2019 from 2:00 until 5:00 am (2.5 mm) and a heavy rain fall event between 26th July 2019 8:00 pm and 27th July 2019 5:00 am with in total 24.7 mm of precipitation (Fig. S4†). During both events, the dry control plots were covered by plastic covers.

Experimental setup

Plot design. The experimental area (Fig. 1) was divided into nine experimental plots, which were arranged as a Latin square with three different treatments. All plots were of 2.4 m lateral length and were separated by 50 cm stripes to allow access. The plots were shielded by plastic edges to prevent runoff from irrigated plots to dry control plots. Within each plot, an inner sampling area of 2 m × 2 m was defined to exclude boundary effects. Three different plot treatments were realized (Fig. 1): (i) G plots were sprayed with glyphosate and 24 h later G and (ii) W plots were artificially irrigated to simulate a heavy rain fall event after a drought period; (iii) dry plots (D) without any treatment served as controls (see details in Section plot treatment).

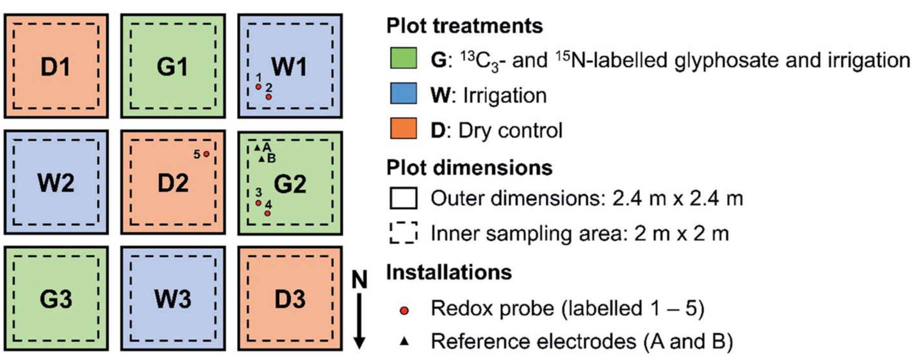

Fig. 1 The experimental area comprised nine plots with three different treatments arranged as a Latin square: dry control plots (D1-3, orange), plots with a simulated heavy rain event (water D_2O-labelled) and two weeks of light irrigation (non-labelled) (W1-3, blue) and plots with a single application of glyphosate one day before the simulated heavy rain event and the two-week irrigation period (G1-3, green). Five redox probes (red circles 1–5) and two reference electrodes (black triangles, A and B, where B served as control for the system) were installed for *in situ* monitoring of redox potentials.

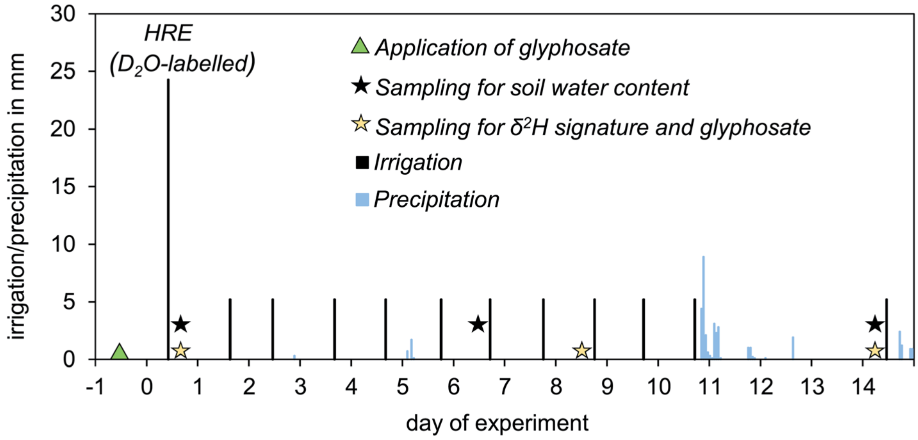

Fig. 2 Irrigation and sampling scheme. The experimental procedure included a 14 day irrigation period with application of glyphosate on particular plots on day −1 (green triangle) and a simulated heavy rain fall event (HRE) with D_2O-labelled irrigation water on day 0 followed by light daily irrigation with non-labelled water (black bars; height indicates irrigation intensity). Natural precipitation is shown as light blue bars; height indicates intensity. Sampling days for different analytes and parameters are labelled with black and golden stars (see legend).

Plot treatment. The irrigation took place from the 16th to 30th July 2019 according to the irrigation scheme in Fig. 2. The day of the simulated heavy rain fall event (16th July) was defined as day 0 of the experiment. All plots were covered one week before the start of the irrigation to secure dry soil conditions at start. One day before the start of the irrigation, 215 mg m^{-2} of $^{13}C_3$–^{15}N-glyphosate, which is equivalent to 2.1 kg ha^{-1}, was sprayed directly onto the soil surface on the G plots (Fig. 1) with a manual sprayer (Schachtner, Ludwigsburg, Germany) to allow sorption equilibration on the soil mineral particles to take place before irrigation. The use of ^{13}C- and ^{15}N-labelled glyphosate allowed to distinguish between potential residual and experiment-associated glyphosate. After 24 h, a heavy rain event (defined by a rain intensity of 7.6 to 50 mm per hour[62]) was simulated on G and W plots by irrigation with 24.3 mm deionized water over approx. 1 hour using watering cans. The irrigation water of the initial heavy rain event was labelled with deuterated water (D_2O) serving as conservative tracer. 2 L of deuterated water (99.9 atom%, Carl Roth, Germany) were added to the irrigation water (1000 L) resulting in an input isotope signature of $\delta^2H = 12\,730‰$ *vs.* Vienna Standard Mean Ocean Water (VSMOW). During the subsequent 10 days and on day 14, 5.2 mm day^{-1} of deionized water without D_2O spike were irrigated as indicated in Fig. 2. The repeated small daily irrigations together with the natural precipitation at days 11–13 were meant to compensate the daily evapotranspiration to maintain closure of large shrinkage cracks.

Samples. Soil cores were collected with a soil corer probe (Humax/GreenGround, Hasle-Rüegsau, Switzerland) and retrieved within PVC liners of 50 cm length and 3.5 cm in diameter. The resulting holes were filled with bentonite. The first soil core samples were taken two hours after the simulated heavy rain event. Samples from all plots were either taken from sites where shrinkage cracks were initially visible directly at the surface or at a distance of at least 25 cm from the next crack in the bulk soil ("sampling type"). The course of the initially present shrinkage cracks was marked before the onset of the experiment, so that in later stages of the experiment, the distinction between the two sampling types was still possible. Of course, even when shrinkage cracks were not visible at the soil surface, shrinkage cracks entering the core from the side may still be present. This issue will be addressed in more detail in the discussion.

Sample labelling included the treatment (glyphosate (G), irrigation (W), dry (D)), where experimental triplicates of each treatment were denoted with numbers 1, 2, and 3, respectively (Fig. 1). Further sample labelling included the "sampling type" (at a shrinkage crack (c) or in bulk soil (b)), and sampling day (T + number) (Table 1). As an example, G1bT3 refers to a sample from the glyphosate + water plot 1 taken from bulk soil at the time point of day 3. GbT3 refers to a set of values from all plot triplicates or the mean value of the three replicate G plots taken from bulk soil on day 3, depending on the context.

For the analysis of soil water δ^2H ratios, cores were taken from W plots along shrinkage cracks and from G plots in bulk soil on days 0, 8, and 14 (WcT0/8/14 and GbT0/8/14). Additionally, soil cores were taken from D plots to determine the natural background δ^2H (DcT0/8/14). The cores of the first day with glyphosate applied (GbT0) were also analyzed for glyphosate in the depth intervals 0–10 cm, 20–30 cm and 40–50 cm. Soil water content was determined on cores from W and D plots

Table 1 Code for samples from the different treatments and of different sampling types

Treatment	
Water + glyphosate	G
Water	W
Dry	D
Sampling type	
Bulk	b
Shrinkage crack	c
Time/day	T + number

taken along shrinkage cracks on days 0, 6 and 14 (W/DcT0/6/14). The analysis of grain size distribution and SOM was conducted on cores DcT10 and WcT14. In the field, all cores were stored on dry ice immediately after sampling to minimize further chemical or microbial processes. Soil cores were stored in the lab at -80 °C until analysis for soil water δ^2H and glyphosate and at -20 °C for grain size and SOM analysis.

***In situ* monitoring of redox potentials.** To monitor *in situ* soil redox potentials, multilevel redox probes (PaleoTerra, Oijen, The Netherlands) were installed on plots of all three treatment types (Fig. 1). The probes had a total length of 1.20 m and consisted of fiberglass and epoxy tubes with in total 10 platinum (Pt) redox electrodes each. For installation, we first drilled a hole with a 2 mm smaller diameter than the probes vertically into the ground. The probes were then installed with the first Pt electrode at a depth of 5 cm, followed by distance intervals of 10 cm below ground level (bgl). Two redox probes each were installed on plots W1 and G2 at a distance of 20 cm (Fig. 1). After installation, the readings of neighboring probes were in close agreement.

Redox potentials were measured as E_{meas} *versus* an Ag/AgCl field reference electrode (KCl sat., $E^0 = 0.199$ V) (PaleoTerra, Oijen, The Netherlands) and are reported *versus* the standard hydrogen electrode (SHE, $E^0 = 0$ V, by definition) as $E_{in\ situ} = E_{\text{meas}} + E_{\text{KCl,sat}}$. Measurement precision was $\pm$ 10 mV. A temperature correction was not done as (i) no conversion factors were available for the used reference electrodes and (ii) the estimated change of E_{meas} due to temperature correction lies within the measurement precision and is thus small compared to the expected redox dynamics. Redox probes and electrodes were connected to a CR 1000 data logger (Campbell Scientific, Logan, Utah, US), which logged redox potentials in all depths every 30 minutes throughout the experiment. A second reference electrode was installed as control for the system ($E_{\text{meas}} = 0 \pm 10$ mV). We will use the following ranges of measured E_{H} values to characterize the overall soil redox conditions at circumneutral pH: oxidizing (>400 mV *vs.* standard hydrogen electrode), weakly reducing (400 to 200 mV), moderately reducing (200 to -100 mV) and strongly reducing (<-100 mV).[64]

Soil water content. Soil water content was determined gravimetrically for irrigated (W) and dry (D) plots. To this end, soil cores were taken vertically including shrinkage cracks and cut in 5 cm intervals between 0 and 10 cm and in 10 cm intervals between 10 and 50 cm. Samples were weighed in triplicates of approx. 1.500 g (scale accuracy $\pm$ 0.001 g) of field wet soil samples before and after freeze drying. The ratio of water loss to dry sample (g g^{-1}) was calculated. The soil water content on the dry plots on day 0 was used as a reference and denoted "initial soil water content".

δ^2H signature of soil water. Hydrogen (δ^2H) isotope ratios of pore water in the soil samples were measured using H_2O(liquid)–H_2O(vapor) pore water equilibration and cavity ring-down laser spectroscopy (Picarro L2120-i and L2130-i).[65] Frozen core samples (50 cm, core set GbT0) were sliced into 10 cm segments. For the determination of the background value the uppermost core segment (0–10 cm) was ignored as it could have been impacted by wind drift of water vapor from irrigation water or evaporation. Aliquots of these segments (between 66 g and 170 g) were placed into 1 L Ziploc freezer bags, double-bagged and stored at 4 °C in the laboratory. For analysis, the bags were filled with dry air in the laboratory and equilibrated for three days before analysis. This allows establishing an isotopic equilibrium between the water vapor in the headspace and the pore water. Three sample measurements were bracketed by the analysis of internal standards for calibration (10 mL water with known isotopic composition in freezer bags). Each sample analysis was monitored over up to 30 min until reaching a plateau indicating stable conditions within the expected measurement accuracy. The average value for the last minute was recorded with stable conditions defined as standard deviations of < 1.0‰ for δ^2H for water vapor. Stable isotope values are given as isotope ratios relative to the international standard (VSMOW) and presented in the delta notation in ‰, $\delta‰ = ((R_{\text{sample}} - R_{\text{standard}})/R_{\text{standard}}) * 1000$, with R as the isotope ratio ^{2}H/^{1}H of the sample and the VSMOW standard.

Afterwards, the samples were dried at 105 °C until reaching constant weight to determine the soil water content gravimetrically comparing wet and dry weight. These data on gravimetric water content were also used for the following mass balance calculations.

Water mass balance. A water mass balance approach was used to calculate the contributions of the 24.3 mm heavy rain event to the profiles and individual depth segments assuming that the irrigation mainly causes a change in water storage and leaching below 50 cm is small. The amount of water volume change was calculated from measured gravimetric water contents in cores from W, G and D plots on day 0 which were converted into volumetric water contents from known bulk densities in the individual depths. This will give the increase in water content and thus the increase in water volume per m^2 due to irrigation compared to the dry plots.

Isotope mass balance. To assess the penetration depth of the irrigation water 2 hours after the irrigation event, the fraction of D_2O-labelled irrigation water in soil water present in every *i*-th depth segment ($f_{\text{IW},zi}$), was calculated using a two-component mixing approach (*i.e.*, isotope mass balance):

$$f_{\text{IW},z_i} = \frac{\delta^2\text{H}_{z_i} - \delta^2\text{H}_0}{\delta^2\text{H}_{\text{IW}} - \delta^2\text{H}_0}$$

where $\delta^2\text{H}_{Zi}$ is the measured isotope ratio of the *i*-th depth segment, $\delta^2\text{H}_{\text{IW}}$ is the isotope ratio of the heavy rain event water (*i.e.*, 12 730‰) and $\delta^2\text{H}_0$ is the background average δ^2H before irrigation (*i.e.*, -47.6‰). The amount of irrigation water in each depth interval was calculated from the water volume $V_{\text{w},zi}$ multiplied by the fraction of irrigation water $f_{\text{IW},zi}$ in each depth interval. The water volume was determined from the analysis of the gravimetric water content in the cores used for isotope analysis converted to volumetric water contents from known bulk densities in the different depths.

Elemental composition. Si and Al were extracted by alkaline fusion.[66] Ca, Cu, Fe, K, Mg, Mn, P were extracted by aqua regia dissolution.[67] Concentrations were determined from the extraction solutions by ICP-OES (inductively coupled plasma optical emission spectroscopy, Agilent 5110, Waldbronn, Germany).

Glyphosate analysis. For glyphosate analysis, representative subsamples from the respective core sections (see Section sampling and sample handling) were lyophilized. Two aliquots of 200 mg of the freeze-dried sample were extracted with 0.5 mL of an aqueous solution containing 50 mM of both NaOH and Na_2HPO_4 upon sonication. This newly developed method is described in detail elsewhere.[68] To quantitate $^{13}C_3$,^{15}N-glyphosate applied to the field, non-labelled glyphosate was added to the extraction medium with a final concentration of 0.5 mg L^{-1}. After centrifugation, the supernatant was acidified with aqueous HCl, precipitates were removed by centrifugation and the remaining supernatant was injected for analysis by capillary electrophoresis-mass spectrometry.[69] Separation was accomplished on a bare fused silica capillary of 68 cm length at a separation voltage of −30 kV and 50 mbar pressure. The background electrolyte was made of 175 mM formic acid titrated to pH 2.8 with ammonia; mass spectrometric detection was accomplished in negative ionization mode. The final concentration was calculated using the signal area of labelled and non-labelled glyphosate considering the liquid–solid ratio (0.5 mL: 200 mg) chosen for the extraction.

Results and discussion

Shrinkage crack dynamics and water infiltration pattern

The soil at the studied field site showed numerous shrinkage cracks with a horizontal extension of several decimeters and widths of up to 1 to 2 cm at the start of the experiment (Fig. 3A) due to a dry period over several weeks before the onset of our experiment. Such shrinkage cracks form preferential flow paths enabling fast and deep penetration of irrigation water, solutes, and fine particles.[11,12,16] Major structural heterogeneities and thus preferential flow paths at the soil surface were absent from day 3 on (Fig. 3B). The high clay content of the soil resulting from weathering of Keuper bedrocks[56,57] led to a fast swelling of the soil upon wetting. During the following days, only narrow surficial shrinkage cracks formed during daytime as a result of the diurnal evaporation (Fig. 3C). These small shrinkage cracks closed rapidly after the daily irrigations of 5.2 mm (Fig. 3D).

The dynamics of opening and closure of shrinkage cracks are in line with the gravimetric soil water content, which we determined for days 0, 6 and 14 from soil cores taken along shrinkage cracks comparing results for irrigated (W and G) and

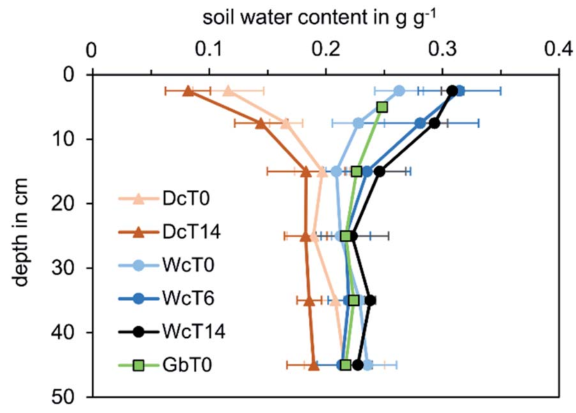

Fig. 4 Gravimetric soil water contents from spots where shrinkage cracks were visible at the surface of dry plots (Dc, orange triangles) and irrigated plots (Wc, blue circles) and from bulk soil from glyphosate + irrigated plots (Gb, green squares) from days 0, 6 and 14 (T0, T6 and T14, respectively) are displayed. Standard deviations were calculated from plot triplicates. The color code from light to dark colors represents the time course of the experiment from day 0 to day 14.

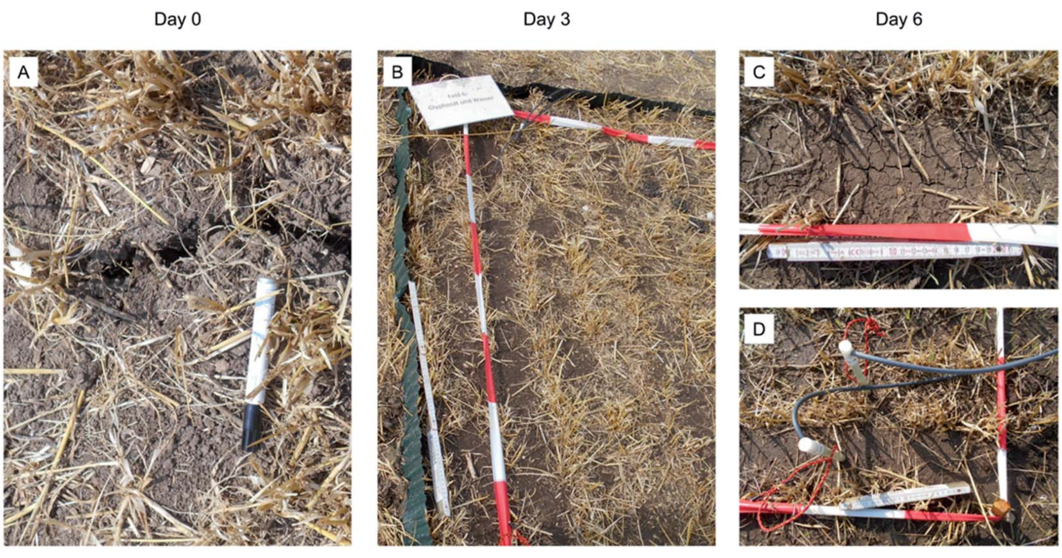

Fig. 3 Dynamics of shrinkage cracks during the experiment. (A) Shrinkage cracks (approx. 1 cm wide) on day 0 at a dry control plot. (B) Disappearance of surficial shrinkage cracks on day 3, following the irrigation scheme in Fig. 2 (length of red bands of the barrier tape is 20 cm). (C and D) Recurring formation of narrow shrinkage cracks (few mm wide) during daytime due to high water evaporation rates (panel C) and immediate closure of these cracks (panel D) during daily 5.2 mm irrigation events on day 6 (length of metering rule is 24 cm).

dry (D) plots (Fig. 4): due to the drought period preceding the irrigation experiment, the initial gravimetric soil water content was low throughout the profile as shown in Fig. 4 exemplarily for the dry plots on day 0 (dataset DcT0), with lowest values (0.12 ± 0.03 g g^{-1}) in the top 5 cm. The water content increased rapidly with depth and was almost twice the value of the top interval in the section 10–20 cm and below reaching 0.20 ± 0.02 g g^{-1}. On the dry plots the water content further decreased by 0.03–0.04 g g^{-1} until day 14 (DcT14) at all observed depths due to high temperatures and evapotranspiration.

On the irrigated plots, the gravimetric water content in the uppermost section (0–5 cm) was 0.26 ± 0.02 g g^{-1} on day 0 (WcT0) and thus more than twice the initial soil water content (DcT0). Water content increased by 1/3 compared to DcT0 in the following section 5–10 cm (0.23 ± 0.02 g g^{-1}) due to heavy irrigation on day 0 (Fig. 4). Subsequent light daily irrigation (5.2 mm day^{-1}) slightly increased the water content of the uppermost section to 0.31 g g^{-1} ± 0.01 g g^{-1} until day 6 (WcT6) when it reached a stable value despite continued light daily irrigation (WcT14). The G plots (GbT0) showed a similar behavior (Fig. 4). The strong increase in water content induced swelling of the soil and closure of the shrinkage cracks (Fig. 3). Generally, on W and G plots the water content gradually decreased with depth within the upper 10 cm. In deeper layers (20 to 50 cm), water content increased by 0.02–0.04 g g^{-1} (WcT0 and GbT0) following the heavy irrigation on day 0 compared to the initial soil water content (DcT0) but then remained nearly constant during the experiment (WcT6/14) (0.22 ± 0.01 g g^{-1}).

The D_2O spike of the irrigation water allowed us to assess the water distribution along the soil profile and to evaluate heterogeneities of the infiltration pathways both in bulk soil and along shrinkage cracks. Note, that the background δ^2H value is −47.6‰ on average. The mean (n = 3) δ^2H isotope values (Fig. 5) illustrated the progression of the irrigation water during the experiment. On day 0, right after the first irrigation pulse, highest δ^2H ratios were present in the top layer, but elevated δ^2H values also were recorded through the entire soil profile. Throughout the irrigation experiment (day 8 and 14), isotope ratios decreased in the top layers whereas the δ^2H values in the deeper layers of the profile were still clearly showing a labelling (Fig. 5A–C). The standard deviations of the averaged (n = 3) δ^2H values decreased from day 0 to day 14 and when comparing results for shrinkage cracks *vs.* bulk samples (Fig. 5). This can be rationalized by the pronounced heterogeneity of the initially dry soil structure due to large shrinkage cracks providing an extended network of preferential flow paths. These preferential flow paths led to an uneven water infiltration during the initial heavy irrigation pulse on the dry soil. Upon closure of the shrinkage cracks (Fig. 3C and D), the distribution of the tracer in both bulk soil and shrinkage crack samples became more uniform (Fig. 5B and C) by mixing of water from the macropore region with the surrounding soil matrix.

The water balance (Fig. S5 A and B†) (considering average values from datasets DcT0 and WcT0/GbT0, Fig. 4) indicated that on W and the G plots 51.8% (*i.e.*, 12.6 mm) and 53.4% (*i.e.*, 13.0 mm) of the 24.3 mm irrigation were present in the segment 0–10 cm, respectively, two hours after the simulated heavy rain event. The small increases of soil water content of 0.02–0.04 g g^{-1} in each of the three 10 cm thick segments between 20 and 50 cm accounted for 12.3% (approx. 3 mm) of the applied irrigation water per depth segment. In total, 95.4% (W) and 95.8% (G) (*i.e.*, 23.2 mm and 23.3 mm) of the irrigation water were recovered in the upper 50 cm two hours after the simulation of the heavy rain event on day 0.

Furthermore, isotope mass balances for individual cores from W and G plots (WcT0 and GbT0) (Fig. S5C and D†) were determined from the δ^2H ratios. Recoveries were 41.5 to 74.4% of the irrigation water (*i.e.*, 10.1 to 18.1 mm) for 4 out of 6 individual soil cores (from plots W1, W3, G1, and G2). Two cores (from plots W2 and G3), however, showed irrigation water recoveries of 238 and 148%, respectively (*i.e.*, 57.9 mm and 36.1 mm). The average recovery from the W plots was 26 ± 27 mm (W_avg, Fig. S5C†) and 22 ± 12 mm (G_avg, Fig. S5D†) from the G plots. These average values closely matched the applied 24.3 mm and thus, on average, the water and the isotope mass balances agreed well. The high standard deviations, however,

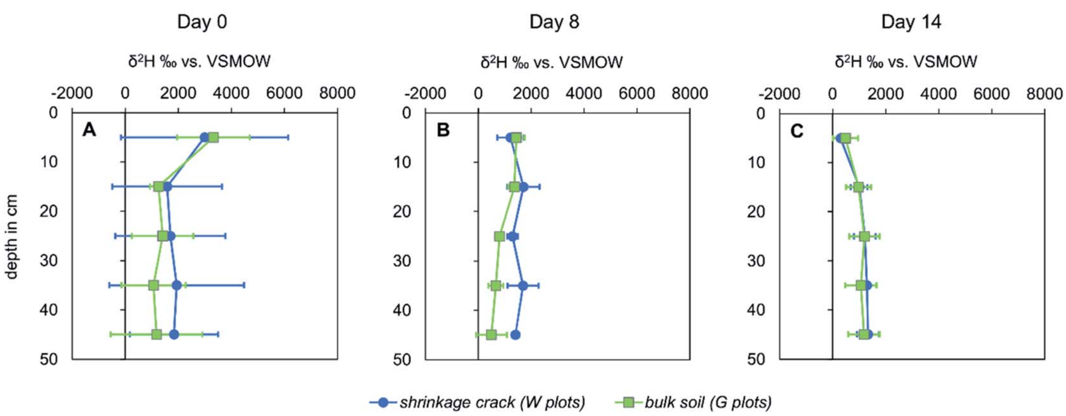

Fig. 5 δ^2H ratios of soil water after the irrigation pulse spiked with D_2O (δ^2H = 12 730‰ *vs.* VSMOW). Mean values and standard deviations of three cores each from irrigated plots (W) sampled along shrinkage cracks (blue circles) and from plots treated with glyphosate (G) sampled in bulk soil (green squares) at days 0, 8 and 14 (Panels A to C). δ^2H ratio in soil water on dry plots (*i.e.*, background before irrigation) was on average −47.6‰ *vs.* VSMOW.

reflect the large differences between the individual plots and thus a substantial heterogeneity of the macropore network below ground. In good agreement with the water mass balances, the mean isotope mass balances W_avg and G_avg (Fig. S5C and D†) also showed that 41–57% (*i.e.*, 10–14 mm) of the irrigation water reached depths of 20 to 50 cm within two hours after the simulated heavy rain event. These numbers also vary strongly between individual soil cores and corroborate the uneven infiltration along preferential flow paths. The small fraction of the irrigation water that is missing in some of the cores likely percolated below the sampling depth of 50 cm or escaped the plots by lateral transport through shrinkage cracks as indicated by the high recoveries of 238% and 148% in cores W2 and G3.

Vertical distribution of glyphosate and D_2O in individual soil cores after the heavy rain event (day 0)

Glyphosate was sprayed manually on selected plots of the field site 24 h prior to the first irrigation pulse. In Fig. 6, we compared glyphosate concentrations (sum of sorbed and dissolved glyphosate) *vs.* fractions of δ^2H-labelled water (*i.e.*, irrigation water) in soil water in %.

The depth distributions of the conservative tracer and the pesticide were similar (Fig. 6A–C and S6†): cores from plots G1 and G2 (bulk soil) showed highest concentrations in the top 10 cm and a steady decrease of concentrations with depth (Fig. 6A and B). In contrast, the core from plot G3 (bulk soil) showed an elevated percentage of irrigation water and elevated concentrations of glyphosate in the subsoil at the bottom of the profile (Fig. 6C). The cores taken from W plots at spots where shrinkage cracks were visible at the surface (WcT0) also showed elevated percentages of D_2O-labelled water in topsoil (W1 and W2) but also in the subsoil (W2 and W3) shortly after the heavy rain event (Fig. 6D).

As glyphosate strongly sorbs to the soil matrix,[27] strong retardation of the compound on the soil surface and in the topsoil is expected when matrix flow is the dominant transport process,[29] both under saturated and unsaturated conditions. However, high glyphosate concentrations occurred in the subsoil already two hours after the first irrigation pulse. The rapid and irregular vertical transport of labelled water together with glyphosate suggests that slow matrix flow through the fine-grained soil was not the major transport mechanism, because transport through advective flow only, would leave glyphosate in the upper few cm (given a porosity of 0.5 it would be the upper 5 cm).

The depth profiles of D_2O-labelled irrigation water in individual soil cores on day 0 parallel glyphosate depth profiles (Fig. 6 and S6†). Together with the high standard deviations of δ^2H values on day 0 (Fig. 5) these findings are consistent with the presence of an extended network of macropores in the subsurface both in soil cores that included shrinkage cracks at the surface and in bulk soil cores. The short time span (two hours) between irrigation and detection of labelled water and glyphosate in the subsoil suggests that the large shrinkage cracks initially present (Fig. 3) served as preferential flow paths for irrigation water and glyphosate into deeper soil levels following the heavy irrigation. The unknown and irregular distribution of macropores below ground hampers the prediction of glyphosate translocation processes at the centimeter scale. These structural heterogeneities, however, levelled out over space (on the meter scale) and time due to closure of the large shrinkage cracks after initial wetting of the soil (Fig. 5).

Although our field data do not provide direct insight into the transport mechanisms of glyphosate, we consider particle-bound transport through the shrinkage crack macropores as the most plausible transport mechanism, because of the strong sorption of glyphosate to soil mineral particles.[27–29,70]

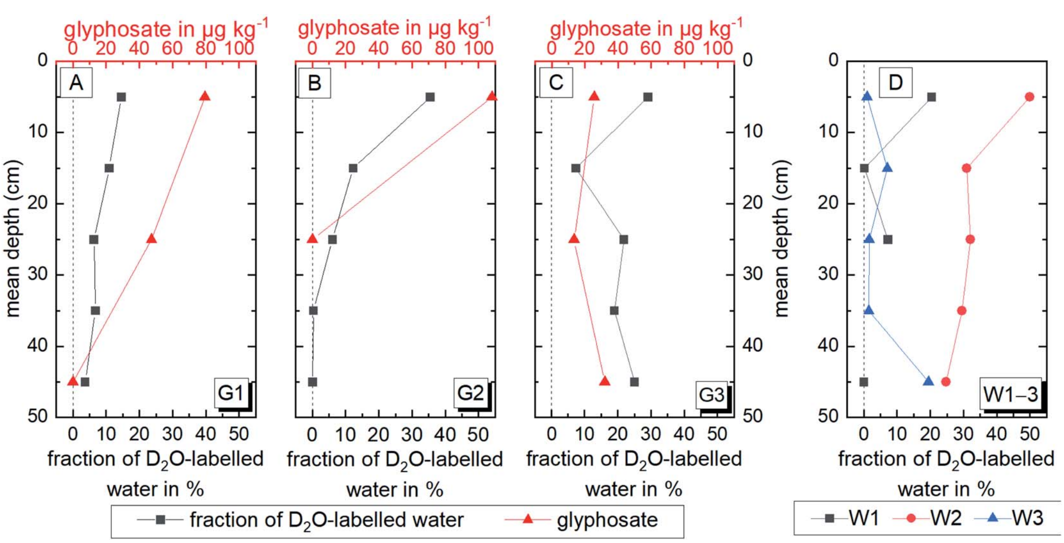

Fig. 6 (A–C) Fraction of D_2O-labelled water in soil water in % (black squares, black horizontal axis) and glyphosate concentrations in µg kg^{-1} (red triangles, red horizontal axis) from plots G1, G2 and G3 (from bulk soil) on day 0 measured in aliquots of the same cores for a direct comparison. (D) (WcT0, but separate data for plots 1–3). Fraction of D_2O-labelled water in soil water in % in cores from all irrigated plots (along shrinkage cracks) on day 0 (W1, black squares; W2, red circles; W3, blue triangles).

Glyphosate leaching from loamy soils triggered by heavy rain events has also been observed in a long-term field study in Denmark.[31–34] Glyphosate transport to depths of at least 1 m was shown by monitoring runoff water from tile drains over 8 months after glyphosate application[31] also suggesting preferential flow as the primary transport mechanism. While the Danish study addressed the leaching of glyphosate under long term wet conditions during the winter in a sandy clay loam,[31] our study showed, by the correlation of D_2O-labelled water and labelled glyphosate, that the rain intensity of a single summer storm was high enough to translocate glyphosate from the soil surface to the subsoil (50 cm depth) even in a dry and more fine grained soil (silty clay loam). The irrigation intensity (24.3 mm per hour) corresponds to rain intensity thresholds determined for the transport of other strongly sorbing pesticides.[71]

Dynamics of redox conditions

Soil redox conditions, which reflect the presence or absence of oxygen and alternative electron acceptors and are a decisive factor for the fate of glyphosate,[50] were monitored *in situ* for all three treatments (W, G, D). Given the neutral soil pH (7.0–7.3), the measured redox potentials correspond to E_H-values at pH7 and further pH corrections were not made. At 5 cm depth on plots W1 and G2, redox potentials around 600 mV with only slight diurnal fluctuations were observed and indicated oxidizing conditions (Fig. 7A, B, S7A and B†). Despite closure of shrinkage cracks upon irrigation, oxygen supply to the uppermost soil layer likely proceeded through the small cracks that opened during the irrigation breaks (Fig. 3C and D). The low amplitude of diurnal variations was partly due to the daily irrigation and temperature changes, as they were also visible on the dry plot albeit to a lower extend than on the irrigated plots (Fig. 7A–C).

Redox potentials rapidly dropped by 400 to 800 mV, *i.e.*, from oxidizing to moderately/strongly reducing conditions on the W1 and G2 plots at 10 and 20 cm depth within a few hours after the irrigation pulse (Fig. 7A, B, S7A and B; † the deviating behaviors of probe 2 at 20 cm will be discussed later). The probe 4 data set (plot G2, Fig. S7B†) did not show such a pronounced drop, but redox potentials clearly responded to the irrigation. Swelling of the clay minerals led to the closure of shrinkage cracks and thus to a reduction of the gas phase. In 30 to 60 cm depth, changes of similar magnitude (400 to 800 mV) were recorded on the W1 plot but with a lag time of 1 to 5 days (Fig. 7D and E) due to the interruption of diffusive oxygen supply from ground surface (Haberer *et al.* (2012)). Clearly, redox conditions changed from oxidizing to moderate or even strongly reducing propagating from top to bottom along the soil profile triggered by the irrigation pulse. Anoxic ground water influence can be excluded given that the sampling was well above capillary fringe, which reached to about 1 m bgl.[54] In general, the response of redox potentials at 30 to 60 cm on the G2 plot was less strong and slower than on W1. It only reached weakly reducing conditions most of the time (see discussion below).

After the initial decrease, redox potential changes in the W1 and G2 plot data sets revealed three domains (1, 2a and 2b) with different characteristics: (1) between 10 cm and 20 cm (partially also at 30 and 40 cm, compare Fig. 7D), redox potential fluctuations between oxidizing and weakly to moderately reducing conditions (amplitudes up to 600 mV) occurred with frequencies of around 24 hours. This diurnal pattern (Fig. 7A and B, 10 cm; Fig. S7A and B,† 10 cm, 20 cm) was likely driven by the daily irrigation and desiccation cycles. Irregular fluctuations of redox potentials on the other hand (Fig. 7B, 20 cm) might be linked to a penetration of oxygen through macropores affected by local shrinkage phenomena, root channels or the redox probe itself. In the second domain (2a) between 30 to 60 cm on plot W1, the

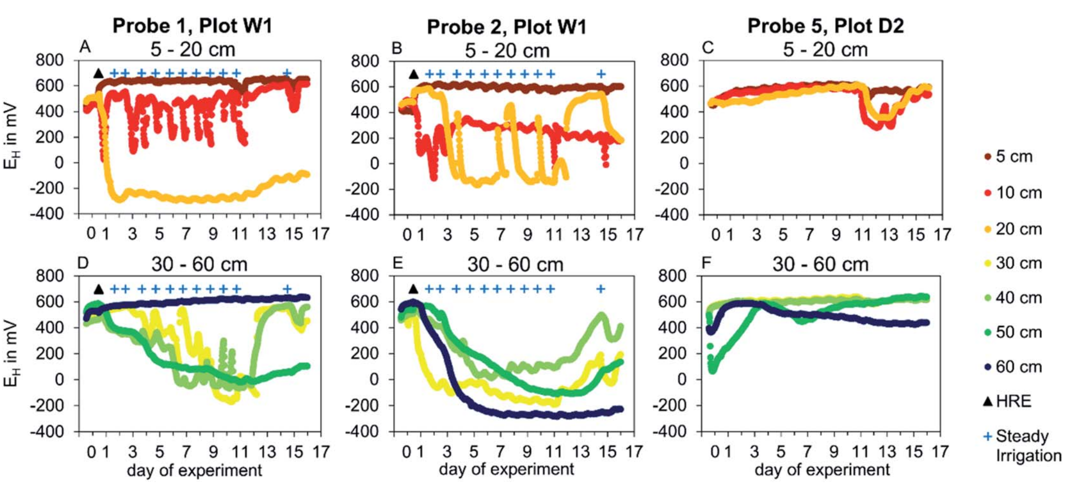

Fig. 7 *In situ* redox potentials at pH7, reported in mV *vs.* SHE. Panels A, B, D, and E show redox potentials measured on a plot treated with water only (W1), panels C and F show potentials measured on a dry control plot (D2). Panels A to C show potentials in the topsoil at depths of 5 cm (brown), 10 cm (light red), and 20 cm (orange). Panels D to F show potentials in the subsoil at depths of 30 cm (yellow), 40 cm (light green), 50 cm (dark green) and 60 cm (blue). The time point of the heavy rain event (HRE) on the W plots is labelled with a black triangle and the time points of the weak (daily) irrigation with dark blue crosses. The redox conditions (E_H-values *vs.* standard hydrogen electrode at pH7) are classified as follows: oxidizing (>400 mV), weakly reducing (400 to 200 mV), moderately reducing (200 to −100 mV) and strongly reducing (<100 mV).[64]

redox potentials after their initial drop to reducing conditions of 0 to −200 mV *vs.* SHE did not show further pronounced fluctuations (Fig. 7D and E). Hence, oxygen exchange with the atmosphere *via* shrinkage cracks was absent. In contrast, in domain 2b, identified at 30 to 60 cm on the G2 plot (Fig. S7C and D†) and at 60 cm on the W1 plot (probe 1, Fig. 7D), the drop of redox potentials was strikingly less pronounced (lowest reading approx. +250 mV; Fig. S7D,† 50 cm and Fig. S7C,† 30 cm) and indicated oxic or weakly reducing conditions most of the time. This behavior suggests that at least parts of the vertical network of macropores were still present so that oxygen supply to the subsoil was not completely interrupted throughout the irrigated plots. As all plots were separated by segments of 50 cm width staying dry, gas exchange through some remaining lateral macropores connections is conceivable. An alternative explanation is, that even if there is enough SOM available for microbial respiration (Fig. S2†), microbial activity in some places in the deeper layers is too low[72–74] to initiate a drop of redox potentials even under oxygen shortage.

The redox-active components in the soil with substantial redox capacity were SOM (Fig. S3†), iron(hydr)oxide minerals and iron-containing clay minerals with a total soil iron content of approx. 2.5% (Table S2†). Both SOM and iron-bearing clay minerals can reversibly take up and release electrons and are electrode active.[42,44] As these compounds can function as geobatteries,[10,75] they might contribute to the dynamics of daily redox potential fluctuations observed in the topsoil. However, the observed diel E_H-changes in the topsoil can be well explained by diel temperature variations.[76] In contrast, the rapid drop of redox potentials after closure of the shrinkage cracks can be rationalized by aerobic microbial respiration using SOM as an electron donor. At that point, the reversible electron donating capacity of geobatteries was exhausted by the extended dry and oxic period prior to the irrigation event.

Redox potentials of the dry D plots (Fig. 7C and F) down to 40 cm indicated oxidizing conditions throughout the soil columns with only small changes during the experiment as expected. One exception was the sudden drop of E_H values at 10 and 20 cm between days 11 and 13, in response to a natural summer rainstorm (24.7 mm) on the evening of day 10 reaching plot D2 as its plastic cover was blown away by strong winds. This incident showed that also a natural heavy rain event induces a response of the redox conditions in 10 and 20 cm depth. The redox conditions in deeper soil layers (30 cm and below) did not react to the rain.

Overall, the redox data demonstrate that irrigation (artificial or natural) induced a drop of redox potentials, although the magnitude of the drop differed between different spots. On the dry plots without irrigation, the atmospheric oxygen supply was unaffected and oxidizing conditions prevailed throughout the entire experiment.

Consequences of dynamic redox conditions in subsoils for sorption and bioavailability of glyphosate

The highly dynamic redox potential profiles as well as the presence of deuterated irrigation water and glyphosate in the subsoil suggest that in the beginning of the experiment the pronounced spatial heterogeneities in the soil in the form of shrinkage cracks reached deep into the subsoil, thereby acting as transport paths for water, glyphosate and oxygen. Once glyphosate had entered the subsoil, it could be exposed to changes in soil redox conditions which can affect its fate. At low redox potentials, microbial or abiotic reduction of Fe(III) to Fe(II) may partially dissolve Fe(III) minerals.[77] Dissolution of such strong sorbents for glyphosate[27,29] will increase its dissolved fraction and thus should increase its mobility and bioavailability in anoxic soils. Enhanced desorption of glyphosate in anoxic compared to oxic soil incubations was reported but glyphosate desorption did not enhance biodegradation.[50] While biotransformation of glyphosate to AMPA or sarcosine under oxic conditions in the field has often been reported,[78,79] anoxic conditions strongly retard[50] or even impede[51] glyphosate biodegradation. Increased desorption in combination with inhibited biodegradation could increase persistence of the compound in the subsoil and could also impose the risk of glyphosate leaching into groundwater or – *via* tile drains[31–33] – into surface water bodies.

Conclusions

In the light of increasing incidents of extreme drought and precipitation events due to climate change, it is imperative to understand their influence on the fate of agrochemicals in highly structured soils. The experimental approach presented here enabled us to study the effects of heavy rainfall after a drought period on the mobility of glyphosate and the redox dynamics in a clayey floodplain soil. By applying glyphosate together with deuterated water as conservative tracer in a field experiment in combination with time resolved *in situ* redox potential measurements the spatial and temporal patterns of water infiltration and pesticide transport as well as the concomitant changes of the redox milieu under field conditions were revealed.

Shrinkage cracks proved to be the key structural soil features controlling physical and biogeochemical processes including water infiltration and redox conditions. Our results demonstrate that shrinkage cracks in dry soils can serve as effective transport paths for atmospheric oxygen, water and even strongly sorbing pesticides like glyphosate. The rain intensity of a typical summer storm event (approx. 25 mm in 1 h) was sufficient to transport water and the model pesticide glyphosate to the subsoil (50 cm) before closure of the shrinkage crack network through swelling of the clay. Further light irrigation resulted in an interruption of the oxygen supply to the depth, which entailed the decrease of soil redox potentials to strongly reducing conditions that potentially impair biodegradation of glyphosate. Given its high application rates worldwide, future research should focus on the fate of glyphosate under highly variable redox conditions.

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Data availability

The data presented in this paper are published in the database of the project "CAMPOS–Catchments as Reactors" at the University of Tübingen, available online (https://hdl.handle.net/10900.1/6aebfaae-6e19-4b3f-81c6-92d4454b6ede).

Author contributions

JS, SH, CG, BW, LC, JW, CH, CP, HP, and EK designed the experiment. JS planned and organized the field work. JS and SH interpreted the data and wrote the manuscript draft. JS, BW, LC, JW, CG and CP conducted the field experiment and collected the samples. LC and CS collected the δ^2H data and CS and JS calculated the mass balances. BW determined the glyphosate concentrations, discussed them with CH and created Fig. 6. JS collected the data on *in situ* redox potentials and contributed the ESI† on soil texture and SOM. JW contributed the data on the soil elemental composition. All authors contributed to writing and reviewing the manuscript.

Conflicts of interest

There are no conflicts to declare.

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References

1 H. O. Venterink, F. Wiegman, G. E. M. Van der Lee and J. E. Vermaat, Role of Active Floodplains for Nutrient Retention in the River Rhine, *J. Environ. Qual.*, 2003, **32**, 1430–1435, DOI: 10.2134/jeq2003.1430.

2 G. J. Lair, F. Zehetner, M. Fiebig, M. H. Gerzabek, C. A. M. van Gestel, T. Hein, S. Hohensinner, P. Hsu, K. C. Jones, G. Jordan, A. A. Koelmans, A. Poot, D. M. E. Slijkerman, K. U. Totsche, E. Bondar-Kunze and J. A. C. Barth, How do long-term development and periodical changes of river–floodplain systems affect the fate of contaminants? results from European rivers, *Environ. Pollut.*, 2009, **157**, 3336–3346, DOI: 10.1016/j.envpol.2009.06.004.

3 A. J. Coby, F. Picardal, E. Shelobolina, H. Xu and E. E. Roden, Repeated anaerobic microbial redox cycling of iron, *Appl. Environ. Microbiol.*, 2011, **77**, 6036–6042, DOI: 10.1128/AEM.00276-11.

4 D. S. Baldwin and A. M. Mitchell, The effects of drying and re-flooding on the sediment and soil nutrient dynamics of lowland river–floodplain systems: a synthesis, *Regulated Rivers: Research & Management*, 2000, **16**, 457–467, DOI: 10.1002/1099-1646(200009/10)16:5<457::AID-RRR597>3.0.CO;2-B.

5 M. Hefting, J. C. Clément, D. Dowrick, A. C. Cosandey, S. Bernal, C. Cimpian, A. Tatur, T. P. Burt and G. Pinay, Water table elevation controls on soil nitrogen cycling in riparian wetlands along a European climatic gradient, *Biogeochemistry*, 2004, **67**, 113–134, DOI: 10.1023/B:BIOG.0000015320.69868.33.

6 D. Coumou and S. Rahmstorf, A decade of weather extremes, *Nat. Clim. Change*, 2012, **2**, 491–496, DOI: 10.1038/nclimate1452.

7 R. M. Horton, J. S. Mankin, C. Lesk, E. Coffel and C. Raymond, A Review of Recent Advances in Research on Extreme Heat Events, *Curr. Clim. Change Rep.*, 2016, **2**, 242–259, DOI: 10.1007/s40641-016-0042-x.

8 S. E. Perkins, L. V. Alexander and J. R. Nairn, Increasing frequency, intensity and duration of observed global heatwaves and warm spells, *Geophys. Res. Lett.*, 2012, **39**, DOI: 10.1029/2012GL053361.

9 T. Borch, R. Kretzschmar, A. Kappler, P. V. Cappellen, M. Ginder-Vogel, A. Voegelin and K. Campbell, Biogeochemical Redox Processes and their Impact on Contaminant Dynamics, *Environ. Sci. Technol.*, 2010, **44**, 15–23, DOI: 10.1021/es9026248.

10 S. Peiffer, A. Kappler, S. B. Haderlein, C. Schmidt, J. M. Byrne, S. Kleindienst, C. Vogt, H. H. Richnow, M. Obst, L. T. Angenent, C. Bryce, C. McCammon and B. Planer-Friedrich, A biogeochemical–hydrological framework for the role of redox-active compounds in aquatic systems, *Nat. Geosci.*, 2021, **14**, 264–272, DOI: 10.1038/s41561-021-00742-z.

11 K. Beven and P. Germann, Macropores and water flow in soils revisited, *Water Resour. Res.*, 2013, **49**, 3071–3092, DOI: 10.1002/wrcr.20156.

12 N. Jarvis, J. Koestel and M. Larsbo, Understanding Preferential Flow in the Vadose Zone: Recent Advances and Future Prospects, *Vadose Zone J.*, 2016, **15**, vzj2016.2009.0075, DOI: 10.2136/vzj2016.09.0075.

13 K. Beven, Kinematic subsurface stormflow, *Water Resour. Res.*, 1981, **17**, 1419–1424, DOI: 10.1029/WR017i005p01419.

14 J. Bouma, Comment on "Micro-, Meso-, and Macroporosity of Soil", *Soil Sci. Soc. Am. J.*, 1981, **45**, 1244–1245, DOI: 10.2136/sssaj1981.03615995004500060050x.

15 M. Flury, H. Flühler, W. A. Jury and J. Leuenberger, Susceptibility of soils to preferential flow of water: a field study, *Water Resour. Res.*, 1994, **30**, 1945–1954, DOI: 10.1029/94WR00871.

16 J. R. Nimmo, The processes of preferential flow in the unsaturated zone, *Soil Sci. Soc. Am. J.*, 2021, **85**, 1–27, DOI: 10.1002/saj2.20143.

17 J. Skopp, Comment on "Micro-, Meso-, and Macroporosity of Soil", *Soil Sci. Soc. Am. J.*, 1981, **45**, 1246, DOI: 10.2136/sssaj1981.03615995004500060052x.

18 J. Bouma and L. W. Dekker, A case study on infiltration into dry clay soil I. Morphological observations, *Geoderma*, 1978, **20**, 27–40, DOI: 10.1016/0016-7061(78)90047-2.

19 M. A. Hardie, W. E. Cotching, R. B. Doyle, G. Holz, S. Lisson and K. Mattern, Effect of antecedent soil moisture on preferential flow in a texture-contrast soil, *J. Hydrol.*, 2011, **398**, 191–201, DOI: 10.1016/j.jhydrol.2010.12.008.

20 J. R. Nimmo, Preferential flow occurs in unsaturated conditions, *Hydrol. Processes*, 2012, **26**, 786–789, DOI: 10.1002/hyp.8380.

21 K. Beven, D. Zhang and A. Mermoud, On the Value of Local Measurements for Prediction of Pesticide Transport at the Field Scale, *Vadose Zone J.*, 2006, **5**, 222–233, DOI: 10.2136/vzj2005.0016.

22 H. Ciglasch, W. Amelung, S. Totrakool and M. Kaupenjohann, Water flow patterns and pesticide fluxes in an upland soil in northern Thailand, *Eur. J. Soil Sci.*, 2005, **56**, 765–777, DOI: 10.1111/j.1365-2389.2005.00712.x.

23 M. Flury, Experimental Evidence of Transport of Pesticides through Field Soils—A Review, *J. Environ. Qual.*, 1996, **25**, 25–45, DOI: 10.2134/jeq1996.00472425002500010005x.

24 R. Kasteel, F. Schnitzler, A. E. Berns, J. Vanderborght and H. Vereecken, Visualization of transport pathways for organic compounds in undisturbed soil monoliths, *Geoderma*, 2013, **195–196**, 70–78, DOI: 10.1016/j.geoderma.2012.11.014.

25 R. P. Schwarzenbach, P. M. Gschwend and D. M. Imboden, *Environmental Organic Chemistry* Wiley, Hoboken, New Jersey, 3rd edn., 2017.

26 M. H. Larsson and N. J. Jarvis, Quantifying interactions between compound properties and macropore flow effects on pesticide leaching, *Pest Manage. Sci.*, 2000, **56**, 133–141, DOI: 10.1002/(SICI)1526-4998(200002)56:2<133::AID-PS103>3.0.CO;2-N.

27 A. L. Gimsing and O. K. Borggaard, Competitive adsorption and desorption of glyphosate and phosphate on clay silicates and oxides, *Clay Miner.*, 2002, **37**, 509–515, DOI: 10.1180/0009855023730049.

28 L. Gulu Zada, PhD dissertation, Eberhard Karls Universität Tübingen, 2019.

29 P. Sprankle, W. F. Meggitt and D. Penner, Adsorption, Mobility, and Microbial Degradation of Glyphosate in the Soil, *Weed Sci.*, 1975, **23**, 229–234, DOI: 10.1017/S0043174500052929.

30 O. K. Borggaard and A. L. Gimsing, Fate of glyphosate in soil and the possibility of leaching to ground and surface waters: a review, *Pest Manage. Sci.*, 2008, **64**, 441–456, DOI: 10.1002/ps.1512.

31 J. Kjær, V. Ernstsen, O. H. Jacobsen, N. Hansen, L. W. de Jonge and P. Olsen, Transport modes and pathways of the strongly sorbing pesticides glyphosate and pendimethalin through structured drained soils, *Chemosphere*, 2011, **84**, 471–479, DOI: 10.1016/j.chemosphere.2011.03.029.

32 J. Kjær, P. Olsen, M. Ullum and R. Grant, Leaching of Glyphosate and Amino-Methylphosphonic Acid from Danish Agricultural Field Sites, *J. Environ. Qual.*, 2005, **34**, 608–620, DOI: 10.2134/jeq2005.0608.

33 T. Norgaard, P. Moldrup, T. Ferré, P. Olsen, A. Rosenbom and L. de Jonge, Leaching of Glyphosate and Aminomethylphosphonic Acid from an Agricultural Field over a Twelve-Year Period, *Vadose Zone J.*, 2014, **13**, DOI: 10.2136/vzj2014.05.0054.

34 A. E. Rosenbom, P. Olsen, F. Plauborg, R. Grant, R. K. Juhler, W. Brüsch and J. Kjær, Pesticide leaching through sandy and loamy fields – Long-term lessons learnt from the Danish Pesticide Leaching Assessment Programme, *Environ. Pollut.*, 2015, **201**, 75–90, DOI: 10.1016/j.envpol.2015.03.002.

35 W. W. Stone and J. T. Wilson, Preferential Flow Estimates to an Agricultural Tile Drain with Implications for Glyphosate Transport, *J. Environ. Qual.*, 2006, **35**, 1825–183, DOI: 10.2134/jeq2006.0068.

36 C. M. Haberer, M. Rolle, O. A. Cirpka and P. Grathwohl, Oxygen Transfer in a Fluctuating Capillary Fringe, *Vadose Zone J.*, 2012, **11**, vzj2011.0056, DOI: 10.2136/vzj2011.0056.

37 M. Alexander, *Biodegradation and Bioremediation*, Academic Press, San Diego, CA, 2 edn., 1999.

38 H.-P. Blume, G. W. Brümmer, H. Fleige, R. Horn, E. Kandeler, I. Kögel-Knabner, R. Kretzschmar, K. Stahr and B.-M. Wilke, *Scheffer/Schachtschabel Soil Science*, Springer, Berlin, Heidelberg, Berlin, Heidelberg, 2016.

39 M. P. Lau, M. Sander, J. Gelbrecht and M. Hupfer, Solid phases as important electron acceptors in freshwater organic sediments, *Biogeochemistry*, 2015, **123**, 49–61, DOI: 10.1007/s10533-014-0052-5.

40 P. G. Tratnyek, T. J. Grundl and S. B. Haderlein, *Aquatic Redox Chemistry*, American Chemical Society, Washington D.C., 2011.

41 M. Usman, J. M. Byrne, A. Chaudhary, S. Orsetti, K. Hanna, C. Ruby, A. Kappler and S. B. Haderlein, Magnetite and Green Rust: Synthesis, Properties, and Environmental Applications of Mixed-Valent Iron Minerals, *Chem. Rev.*, 2018, **118**, 3251–3304, DOI: 10.1021/acs.chemrev.7b00224.

42 M. Aeschbacher, M. Sander and R. P. Schwarzenbach, Novel Electrochemical Approach to Assess the Redox Properties of Humic Substances, *Environ. Sci. Technol.*, 2010, **44**, 87–93, DOI: 10.1021/es902627p.

43 M. Byrne James, N. Klueglein, C. Pearce, M. Rosso Kevin, E. Appel and A. Kappler, Redox cycling of Fe(II) and Fe(III) in magnetite by Fe-metabolizing bacteria, *Science*, 2015, **347**, 1473–1476, DOI: 10.1126/science.aaa4834.

44 C. A. Gorski, L. Klüpfel, A. Voegelin, M. Sander and T. B. Hofstetter, Redox Properties of Structural Fe in Clay Minerals. 2. Electrochemical and Spectroscopic

Characterization of Electron Transfer Irreversibility in Ferruginous Smectite, SWa-1, *Environ. Sci. Technol.*, 2012, **46**, 9369–9377, DOI: 10.1021/es302014u.

45 M. Aeschbacher, D. Vergari, R. P. Schwarzenbach and M. Sander, Electrochemical Analysis of Proton and Electron Transfer Equilibria of the Reducible Moieties in Humic Acids, *Environ. Sci. Technol.*, 2011, **45**, 8385–8394, DOI: 10.1021/es201981g.

46 M. Sander, T. B. Hofstetter and C. A. Gorski, Electrochemical Analyses of Redox-Active Iron Minerals: A Review of Nonmediated and Mediated Approaches, *Environ. Sci. Technol.*, 2015, **49**, 5862–5878, DOI: 10.1021/acs.est.5b00006.

47 W. H. Patrick, R. P. Gambrell and S. P. Faulkner, in *Methods of Soil Analysis*, ed. D. L. Sparks, 1996, DOI: https://doi.org/DOI: 10.2136/sssabookser5.3.c42, pp. 1255–1273.

48 S. Peiffer, O. Klemm, K. Pecher and R. Hollerung, Redox measurements in aqueous solutions—a theoretical approach to data interpretation, based on electrode kinetics, *J. Contam. Hydrol.*, 1992, **10**, 1–18, DOI: 10.1016/0169-7722(92)90041-C.

49 Z. Zhang and A. Furman, Soil redox dynamics under dynamic hydrologic regimes – a review, *Sci. Total Environ.*, 2021, **763**, 143026, DOI: 10.1016/j.scitotenv.2020.143026.

50 R. G. Kanissery, A. Welsh and G. K. Sims, Effect of Soil Aeration and Phosphate Addition on the Microbial Bioavailability of Carbon-14-Glyphosate, *J. Environ. Qual.*, 2015, **44**, 137–144, DOI: 10.2134/jeq2014.08.0331.

51 S. R. Sørensen, A. Schultz, O. S. Jacobsen and J. Aamand, Sorption, desorption and mineralisation of the herbicides glyphosate and MCPA in samples from two Danish soil and subsurface profiles, *Environ. Pollut.*, 2006, **141**, 184–194, DOI: 10.1016/j.envpol.2005.07.023.

52 O. F. Geyer, M. P. Gwinner, M. Geyer and D. Ellwanger, *Geologie von Baden-Württemberg*, Schweizerbart, Stuttgart, 5th edn, 2011.

53 P. Grathwohl, H. Rügner, T. Wöhling, K. Osenbrück, M. Schwientek, S. Gayler, U. Wollschläger, B. Selle, M. Pause, J.-O. Delfs, M. Grzeschik, U. Weller, M. Ivanov, O. A. Cirpka, U. Maier, B. Kuch, W. Nowak, V. Wulfmeyer, K. Warrach-Sagi, T. Streck, S. Attinger, L. Bilke, P. Dietrich, J. H. Fleckenstein, T. Kalbacher, O. Kolditz, K. Rink, L. Samaniego, H.-J. Vogel, U. Werban and G. Teutsch, Catchments as reactors: a comprehensive approach for water fluxes and solute turnover, *Environ. Earth Sci.*, 2013, **69**, 317–333, DOI: 10.1007/s12665-013-2281-7.

54 S. Martin, S. Klingler, P. Dietrich, C. Leven and O. A. Cirpka, Structural controls on the hydrogeological functioning of a floodplain, *Hydrogeol. J.*, 2020, **28**, 2675–2696, DOI: 10.1007/s10040-020-02225-8.

55 M. Schwientek, K. Osenbrück and M. Fleischer, Investigating hydrological drivers of nitrate export dynamics in two agricultural catchments in Germany using high-frequency data series, *Environ. Earth Sci.*, 2013, **69**, 381–393, DOI: 10.1007/s12665-013-2322-2.

56 Landesamt für Geologie Rohstoffe und Bergbau Baden-Württemberg, *LGRB-kartenviewer - Layer GeoLa GK50 (1:50 000), Geologische Einheiten (Flaechen)*, 2021, https://maps.lgrb-bw.de, (accessed 14.10.2021).

57 *Landesamt für Geologie Rohstoffe und Bergbau Baden-Württemberg, LGRB-Kartenviewer - Layer GeoLa BK50 (1:50 000)*, Bodenkundliche Einheiten, 2021, https://maps.lgrb-bw.de, (accessed 14.10.2021).

58 IUSS Working Group WRB, *World Reference Base for Soil Resources 2014, Update 2015 International Soil Classification System for Naming Soils and Creating Legends for Soil Maps*, FAO, Rome, 2015.

59 DIN Deutsches Institut für Normung e. V., *DIN EN ISO 17892-4 Geotechnical Investigation and Testing - Laboratory Testing of Soil - Part 4: Determination of Particle Size Distribution*, Beuth, Berlin, 2017.

60 German Meteorological Sevice (DWD), *Klimatabelle Temperature/Niederschlag*, Station Stuttgart-Echterdingen, 2021, https://www.dwd.de/DE/leistungen/kvo/baden_wuerttemberg.html?nn=480164, (accessed 02.12.2021).

61 Agrarmeteorologie Baden-Württemberg, Station Unterjesingen, 2021, https://www.wetter-bw.de/Internet/AM/NotesBwAM.nsf/bwweb/4262596897754529c1257ca8002f9d19?OpenDocument, (accessed 14.10.2021).

62 American Meteorological Society, "*Rain*". Glossary of Meteorology, 2021, https://glossary.ametsoc.org/wiki/Rain, (accessed 14.10.2021).

63 R. G. Allen, L. S. Pereira, D. Raes and M. Smith, *FAO Irrigation and Drainage Paper 56 - Crop Evapotranspiration - Guidelines for Computing Crop Water Requirements*, FAO - Food and Agriculture Organization of the United Nations, Rome, 1998.

64 T. Mansfeldt, In situ long-term redox potential measurements in a dyked marsh soil, *J. Plant Nutr. Soil Sci.*, 2003, **166**, 210–219, DOI: 10.1002/jpln.200390031.

65 L. I. Wassenaar, M. J. Hendry, V. L. Chostner and G. P. Lis, High Resolution Pore Water δ^2H and δ18O Measurements by H_2O(liquid)–H_2O(vapor) Equilibration Laser Spectroscopy, *Environ. Sci. Technol.*, 2008, **42**, 9262–9267, DOI: 10.1021/es802065s.

66 DIN Deutsches Institut für Normung e. V., *DIN ISO 14869-2 Soil Quality - Dissolution for the Determination of Total Element Content - Part 2: Dissolution by Alkaline Fusion*, Beuth, Berlin, 2003.

67 VDLUFA, in *Methodenbuch Band VII Umweltanalytik VDLUFA Verband Deutscher Landwirtschaftlicher Untersuchungs- und Forschungsanstalten e.V.*, 2011, vol. 4, p. 690.

68 B. Wimmer, H. Neidhardt, M. Schwientek, S. B. Haderlein and C. Huhn, *Phosphate Addition Enhances Alkaline Extraction of Glyphosate from Highly Sorptive Soils and Aquatic Sediments*, Pest Management Science, 2022, DOI: 10.1002/ps.6883.

69 B. Wimmer, M. Pattky, L. G. Zada, M. Meixner, S. B. Haderlein, H.-P. Zimmermann and C. Huhn, Capillary electrophoresis-mass spectrometry for the direct analysis of glyphosate: method development and application to beer beverages and environmental studies,

Anal. Bioanal. Chem., 2020, **412**, 4967–4983, DOI: 10.1007/s00216-020-02751-0.

70 H. de Jonge, L. W. de Jonge and O. H. Jacobsen, [14C] Glyphosate transport in undisturbed topsoil columns, *Pest Manage. Sci.*, 2000, **56**, 909–915, DOI: 10.1002/1526-4998(200010)56:10<909::AID-PS227>3.0.CO;2-5.

71 G. S. McGrath, C. Hinz, M. Sivapalan, J. Dressel, T. Pütz and H. Vereecken, *Identifying a Rainfall Event Threshold Triggering Herbicide Leaching by Preferential Flow*, Water Resources Research, 2010, vol. 46, DOI: 10.1029/2008WR007506.

72 S. Loeppmann, E. Blagodatskaya, J. Pausch and Y. Kuzyakov, Enzyme properties down the soil profile - A matter of substrate quality in rhizosphere and detritusphere, *Soil Biol. Biochem.*, 2016, **103**, 274–283, DOI: 10.1016/j.soilbio.2016.08.023.

73 R. M. Niemi, M. Vepsäläinen, K. Wallenius, S. Simpanen, L. Alakukku and L. Pietola, Temporal and soil depth-related variation in soil enzyme activities and in root growth of red clover (Trifolium pratense) and timothy (Phleum pratense) in the field, *Appl. Soil Ecol.*, 2005, **30**, 113–125, DOI: 10.1016/j.apsoil.2005.02.003.

74 S. C. Stock, M. Köster, M. A. Dippold, F. Nájera, F. Matus, C. Merino, J. Boy, S. Spielvogel, A. Gorbushina and Y. Kuzyakov, Environmental drivers and stoichiometric constraints on enzyme activities in soils from rhizosphere to continental scale, *Geoderma*, 2019, **337**, 973–982, DOI: 10.1016/j.geoderma.2018.10.030.

75 L. Klüpfel, A. Piepenbrock, A. Kappler and M. Sander, Humic substances as fully regenerable electron acceptors in recurrently anoxic environments, *Nat. Geosci.*, 2014, **7**, 195–200, DOI: 10.1038/ngeo2084.

76 K. Dorau, B. Bohn, L. Weihermüller and T. Mansfeldt, Temperature-induced diurnal redox potential in soil, *Environ. Sci.: Processes Impacts*, 2021, **23**, 1782–1790, DOI: 10.1039/D1EM00254F.

77 R. M. Cornell and U. Schwertmann, *The Iron Oxides: Structure, Properties, Reactions, Occurences and Uses*, Wiley-VCH Verlag GmbH & Co. KGaA Weinheim, 2 edn, 2003.

78 A. V. Sviridov, T. V. Shushkova, I. T. Ermakova, E. V. Ivanova, D. O. Epiktetov and A. A. Leontievsky, Microbial degradation of glyphosate herbicides (Review), *Appl. Biochem. Microbiol.*, 2015, **51**, 188–195, DOI: 10.1134/S0003683815020209.

79 H. Zhan, Y. Feng, X. Fan and S. Chen, Recent advances in glyphosate biodegradation, *Appl. Microbiol. Biotechnol.*, 2018, **102**, 5033–5043, DOI: 10.1007/s00253-018-9035-0.